FINAL TERM MCQS
(BT401)

PREPARE BY: ASHAM

TOPIC NO# 1

1. ABSA stands for ……**access and benefit sharing agreement.**

TOPIC NO# 2

2. PGRFA stands for…….**plant genetic resource for food and agriculture**
3. Animal genetic resource are (Animal genetic resource, Animal genetic resource for food and agriculture, 3.farm animal genetic resource) …….. **all of these**

TOPIC NO# 3

4. Genetic resources are sometimes called_ **first resource**
5. …..are link from generation to generation of all living matter……**genes**

TOPIC NO# 4-5

6. Landraces is a__ **local variety of domesticated plants**
7. ____are varieties developed since the advent of scientific agriculture in the late nineteenth century and that are no longer cultivated…… **Obsolete varieties**
8. The sum total of all the gene present in a crop and its related species constitutes its…… **germplasm**

TOPIC NO#6

9. Conservation of plant genetic resource is necessary for…….. **Food scrutiny and Agro-biodiversity (both A&B)**

TOPIC NO#7

10. Special care unit…… **ex situ**
11. ex-situ conservation include …… **all given (Seed Gene bank, Botanical gardens, zoological parks)**

12. _____Is an example of an ex-situ conservation……**seed bank**
13. Gene banks are also known as…….. **Germplasm banks.**
14. Field gene banks also called plant gene banks are areas of land in which _____collections of growing plants are assembled …….. **germplasm**
15. Gene bank is a type of ……**ex-situ conservation**
16. Gene sanctuaries are the type of…… **in-situ conservation**
17. The natural protected area for the conservation of genetic diversity is called…….. **gene sanctuary**
18. Area of great genetic diversity are protected from human interface are known as ……**gene sanctuary**
19. Which of the following statement truly describe the biosphere reserve……. **All given** (Biosphere reserves are areas of terrestrial and coastal ecosystems promoting solutions to reconcile the conservation of biodiversity with its sustainable use)

TOPIC NO#8

20. _____can be used to introduce species that left an area……. **Ex-situ conversion**

TOPIC NO#9

21. Plant diversity is urgently & effectively conserved is an objective of…….**GSPC**
22. National strategy of PGRFA need to be review if ------- changes ……. **world**

TOPIC NO#12

23. _____is a pioneer species, easily regenerated from seed……. ***Acacia nilotica***

TOPIC NO# 13-14

24. Pakistan in kinow production…….. **Six largest producer**

TOPIC NO#15

25. Peste de petits ruminants (PPR) also known as……..**sheep & goat plague**

TOPIC NO#16-17

26. Sheep was domesticated by humans in……..**10,000BC**

TOPIC NO#18

27. Snow leopard population in Pakistan left **200**

28. Himalayan brown beer population in Pakistan……150-200

TOPIC NO#19

29. Seaweeds population is……13000

30. Which disease is spread due to consuming mercury-poisoned fishes by people …….. MNAMATA disease.

TOPIC NO# 20-21

31. Closely related individuals…….. inbreeding

32. National strategy for PGRFA may help…… Country

33. Some crustaceans migrate for…… mating , breeding

34. Example of extinct species……. Passenger pigeon, white rhino

TOPIC NO# 22

35. NCCP stands for ……. national culture collection of Pakistan

36. Microorganisms are……more diverse

37. ___is defined as the number of species represented in a specific region landscape or an ecological community……..Species richness

TOPIC NO# 24

38. Anthropogenic impact is apparent in the …….. Coastal zone

39. Woody plants like trees have a -- genetic diversity then vascular plants like grasses……..Higher

TOPIC NO# 25-27

40. The Indus River dolphin is one of the world's ……rarest mammals.

TOPIC NO# 28-30

41. GEWIS stand for…… A genome wide interaction scan

42. A _____ can be taken to analyses G across E interactions…… polygenic approach

43. Which of the following approach can be taken to analyze GxE interaction…… polygenic

44. Phenylketonuria caused by…… Mutation

TOPIC NO#31-32

45. Gene pool types ……3

TOPIC NO#33

46. The strongest effect of genetic drift inn…… small population

TOPIC NO#34

47. More genetic variation in populations…….. **natural selection**

48. If the advantage gene is removed then the effect is called…… **genetic drift**

TOPIC NO#35-38

49. Bottleneck effect elephant seal……. **Over hunting**

TOPIC NO#40

50. Mild breeding is type of…….. **inbreeding**

TOPIC NO#42

51. Triggers of migration…… **all (. local climate 2. local availability of food 3. the season of the year 4. for mating reasons)**

TOPIC NO#43

52. Crow follow migration…….. **Daily**

TOPIC NO#45

53. Longest migration was observed in……. **artic tern bird**

54. Wildebeest are the part of ……**great migration**

TOPIC NO#46

55. There are_____types of genome transfer …… **two**

56. _____can be transferred sexually or asexually …… **vertical gene transfer**

TOPIC NO#47-48

57. Darwin finches classical example of……. **Adaptive radiation**

58. _____is an example allopatric speciation……. **Darwin finches**

59. _____often result of reproductive isolation……**Sympatric speciation**

TOPIC NO#50

60. Extinction rate of mammals ……**20%**

TOPIC NO#51

61. Coral reefs are also threatened Reefs are home to ____of marine animal……**25%**

62. Rhino are hunted for……**horns**

MAAH WARRIORS

TOPIC NO#52

63. _____produce by dart poison frog……. **Toxins**

TOPIC NO#55-56

64. Sterculia khasiana was endemic tree of ……**Khasi Hills in Meghalaya in India**
65. Calypha belongs to family_____spurge family ……**euphorbiaceae / spurge family**
66. Sigillaria was a tree ……..**spore bearing**

TOPIC NO#57-58

67. Species that are threatened are sometimes characterized (population dynamics measure of critical dispensation, a mathematical measure of biomass)related to population growth rate…… **all are given**
68. Houghton's goldenrod typically grows on moist (sandy beaches and shallow depressions between low sand ridges along the shoreline)…….. **all are given**

TOPIC NO#59

69. Cryopreservation in liquid nitrogen …… **-196degrees**
70. Cryopreservation is storage of materials at ____ temperature…….. **low**
71. ___ Are preserved by cryopreservation method…….....**genetic material**

TOPIC NO#61

72. _____is a strategy wherein samples of animal genetic materials are preserved cryogenically……. **Cory-conservation of animal genetic resources**
73. Genetic materials that are typically cryogenically preserved include ……**sperm, oocytes, embryos and somatic cells.**
74. FAO stands for……. **Food and agriculture organization**
75. What is Dimethyl sulfoxide used for…… **chelating agent**
76. How long a seed remains viable is known as it is ……. **Viability**

TOPIC NO#63

77. OHSS stand for …….. **Ovarian hyper stimulation syndrome**
78. Thermal gradient can induce_____ due to uneven expansion or contraction in the biochemical system ……. **Mechanical Stress**
79. Cryopreservation helps in the preservation of biological materials……. **Sperm, gametes, embryos**

TOPIC NO#64

80. The 1st report on fish sperm cryopreservation was published by…… **Blaxter (1953).**

TOPIC NO#65

81. Which of the following is the earliest modern protected area? ……..**Yellow stones national park**

TOPIC NO#66

82. The problem in strict nature reserves is (a disease b, climate change, air pollution)……..**Both B & C**

TOPIC NO#68-69

83. WCPA stands for……**world commission on protected areas**

84. CBD stand for……**Convention on Biological Diversity**

85. Natural geological and geomorphological features…… **Waterfalls, cliffs, craters, caves etc.**

86. Which one of the following is culturally influenced in natural reserve…….**Cave Dwelling**

87. They are generally quite small-protected areas and often have high visitor value category VIII…… **National Monument-Feature**

88. _____are small sites they focus on more prominent natural features…… **National Monument-Feature**

TOPIC NO#70-71

89. Which category area are not associated with presence and intervention ……**category-lV**

90. Which category areas are not necessary associated with human presence and intervention……. **Category lV**

91. Main objective of habitat species management area to maintain, conserve and restore species and habitats…… **All of these**

92. Habitat management work…….. **all of the above**

TOPIC NO#72-73

93. ___protected areas act as buffer ……**Category V**

94. Encourage the conservation of agrobiodiversity and aquatic biodiversity……. **Land & seascape**

95. Which of the following category encourage the conservation of agrobiodiversity and aquatic biodiversity…… **Category V**

96. The main objective of category V is…….. **maintain a balance interaction of nature and culture**

TOPIC NO#74-75

97. Large natural areas comes into……**category VI**

98. The category VI is not designed to accommodate……. **large-scale industrial harvest.**

TOPIC NO#76-77

99. Largest national park of Pakistan……**Hingol National Park**

100. Hingol national park declared reserved in ……**1988**

101. Blackbuck became virtually extinct in the Cholistan Desert but the species has been reintroduced in ……. **Lal Suhanra**

102. When species has been reintroduced in Lal suhanra national park..... **Blackbuck**

TOPIC NO#78-79

103. Sir Sadiq Muhammad Khan established……..**Bahawalpur Zoo**

104. Bahawalpur zoo covers……**25 acres**

TOPIC NO#82-84

105. In 2013,_____rasmar sites has been declared in Pakistan……**19**

106. Ramsar conversion was negotiated in ____by countries and NGOs …….. **1960**

TOPIC NO#85-86

107. Since the_____ population of the Indus Dolphin has significantly increased here……**1970**

108. A sanctuary is a protected area which is reserved for the conservation of only animal and human activities like……. **harvesting of timber**

109. Chashma and Tuansa barrage declared wild life sanctuaries by……. **Punjab government**

110. Desert wolf is ____ in Cholistan Wildlife Sanctuary…….. **Rare**

111. Rann of Kutch wildlife sanctuary is part of the…… **the desert**

112. Sarus crane is one of _____ Species at Runn of Kutch Wildlife sanctuary…….**.threatened**

113. Threatened species of Rann of Kutch is…….. **All (Great Indian bustard, Houbara bustard, Sarus crane)**

114. Which of the following statement is correct about the sanctuary…….. **A reserved area meant for preservation and development of endangered species**

TOPIC NO#87

115. If hunting is prohibited, a game reserve may be considered a ……." **Nature Reserve"**

116. _____an area wherein controlled hunting and shooting is permitted on permit basis" is known as……..**game reverse**

117. In the____the, major focus is specifically the animals…….. **game reserves**

118. In the game reserves, the major focus is specifically the……. **animals.**

TOPIC NO#88

119. The Buler's Fruit bat is listed as _____ on the IUCN red list……..**Critically Endangered**

120. The IUCN has listed Great Indian Bustard as_____in 2011…….. **Critically Endangered**

121. Hunting habitat loss and fragmentation have reduce the number of this specie…… **Great Indian bustard**

122. Madagascar Pochard die at young age due to……. **All of them (absence of adequate food items and too deep water)**

TOPIC NO#89

123. IBPGR stand for…….. **International Board for Plant Genetic Resource**

124. In _____conservation of habitats rich in genetic diversity was recommended in the UN conference……. **1972**

125. an International Board for Plant Genetic Resource (IBPGR) was established in……. **1972**

126. Plant seeds are stored in _____for long-term storage……. **In-vivo Gene Bank**

127. Germplasm conservation is important to conserve the ___ plant……. **Wild**

128. Since ____in situ conservation has received high priority in the world conservation strategy……..**1980,**

TOPIC NO#91

129. High humidity increases _____growth…….. **fungal**

130. Vegetative propagation is……. **tissue culture**

131. Tissue culture conservation is an alternative to.....……. **Vegetative propagation.**

132. The removal of the medium and transfer of cells from its previous culture into fresh growth medium for further propagation of the cell is called……. **Sub culturing**

133. During tissue culture growth room temperature range from_____depending on species requirements…… **22°C to 28°C**

134. Suitable light intensity range must of the plant culture is…… **10 to 1000 µmol S-1 m-2.**

135. Light intensity for most plants culture require ……. **50-200 umol-S-1-m-2**

TOPIC NO#92

136. The useful of the specimen for PCR based assays is greater than 4 years when stored at……. **-80 C°**

137. DNA Banks are used for the storage of those species that cannot be conserved in process of conservation…… **in situ**

138. DNA of plant can be preserved by rapid drying of plant samples with the help of ____material…… **silica gel**

139. Lyophilisation during DNA storage process helps to _____ the DNA…… **preserve**

140. _____is the secure long-term storage of an individual's genetic material…… **DNA bank**

141. ____can now be considered as a means of complimentary conservation……. **DNA banks**

142. The first genomic libraries were cloned in ……**Plasmid**

TOPIC NO#93-95

143. Gene sanctuaries provide way to preserve (a. wild species, natural ecosystem. Natural habitat) ……**all of these**

144. Ex-situ conservation is also known as……..**static conservation**

145. In _____gene sanctuaries for conservation of wild relatives of coffee was setup in 1984…… **Ethiopia**

146. Gene sanctuaries not only preserve the existing genetic diversity present in a population due to which new _____ combination would appear with time…….. **Allele and gene**

147. Gene sanctuary is generally established in the centres of ……**diversity**

148. There are two main drawbacks of gene sanctuary one of which is that…… **It cannot be easily maintained**

TOPIC NO#96

149. ____is a relatively cheap method for conserving a broad range of germplasm…… **Seed storage**

TOPIC NO#97

150. Pollinator are_____ major group of invertebrates……**SECOND**

151. Invertebrates accounts for_____% of the animals on earth……**95%**

152. ___ Organisms are described as engineers of ecosystem…… **earthworms**

153. It has been at least _____of world food production comes from crops that are dependent on insect pollination…… **35 percent**

154. At least 35 percent of world food production comes from crops that are dependent on…….. **Insect pollination**.

155. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites help maintain the chemical fertility needed for ……..**plant growth**

156. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites, have been described as ……..**"ecosystem engineers".**

157. Perhaps the most neglected group of all – in research, in farming practices, and in policies and strategies for agriculture and biodiversity – are the soil-dwelling ……**invertebrates**

TOPIC NO#98

158. _____have created barriers to the migration of invertebrate species…… **Human activities**

159. ____ is Key factor for regulating many of bio chemical process …… **temperature**

160. Most invertebrates are expected to change their geographical distribution…… **in response to climate change**

TOPIC NO#99

161. _____help to maintain soil structure and the availability of water throughout the soil profile…… **Earthworms**

162. This lack of diversity may inhibit the ability of the population to respond to …..**climate change.**

163. The capacity of ecosystems to adapt is diminished when biodiversity of_____ is lost…… **invertebrate**

164. The adaptive capacity of ecosystem is more likely to dependent on…… **invertebrate biodiversity**

165. Studies have shown that the presence of these animals can help to……. **alleviate**

TOPIC NO#100

166. _____are needed to be able to cope with changing climatic conditions, and increasing demands for food and retain the capacity to adapt to potential changes in the types of crops grown…….**Pollinator populations**

167. Considering the better studied mammals and birds 100% of the currently described species have been evaluated for their conservation statuses and out of these ____ out of 9,990 birds species are considered to be endangered……**12%**

168. Endemic species are…….. **Species localized in a specific region**

TOPIC NO#101

169. Though forest trees are known for showing great _____in their response to climate changes…… **plasticity**

170. Forest genetic resources or tree genetic resources are _____of shrub and tree species of actual or future value…… **genetic material**

171. In _____the Food and Agriculture Organization of the United Nations published the first State of the World's FGR……**2014**

172. Diversity of forest genetic resources enables the species to adapt to…… **All given (temperature changes, drought, pests, diseases and forest fires)**

173. Forest help in driving……..**Climate Change**

174. Forests are important due to……**all given (stand, population or landscape of trees, and typically other associated woody plants)**

TOPIC NO#102

175. Forests are important due to…… **(All given)**

176. Changes in climate also make it possible for ____and diseases to invade new areas, destroying the forests there…….. **Pests**

177. Forests majorly remove _____from the atmosphere and produce oxygen…….. **Carbon dioxide**

178. In Europe one estimate put the role of value marketed _____ non wood goods from forest at……. **€2.3 billion**

179. Globally, forests remove _____from the atmosphere and produce oxygen…….. **carbon dioxide**

TOPIC NO#103

180. Climate has direct effect on ……**FGR**

181. Tree mortality in ecosystem increase due to change in…….. **all given (extreme climatic events, particularly drought in combination with widespread regeneration failure)**

182. Anthropogenic climate change leading to future large-scale "dieback" in……. **Amazonian rain forest.**

TOPIC NO#104

183. The ability of one genotype to produce more than one phenotypes when exposed to different environment conditions without genetic change is called…….**phenotypic plasticity**

184. _____is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions……. **Phenotypic plasticity**

185. Tree populations rely on three interplaying mechanisms to respond to environmental change…… **(All given) adaptation, Migration, phenotypic plasticity**

186. Tree populations rely on_____ interplaying mechanisms to respond to environmental change…… **three**

TOPIC NO#105

187. Hybridisation may also introduce the new genetic variation required by trees to adapt to…….. **Novel environments.**

188. Some insects use tree terpenes as precursors for their communication_____ or incorporate them into their own defense systems…… **pheromones**

189. Natural selection favors genotypes with …….. **(All given) increased tolerance or resistance to disturbances, and phenotypic plasticity**

190. ___describes a situation where two or more) species reciprocally affect each other's evolution……. **Co-evolution**

TOPIC NO#106

191. Climate change-related traits including……. **(All given) plasticity and adaptation**

192. Candidate genes for drought tolerance include those involved in…….. **(All given) the synthesis of abscisic acid, transcriptional regulators**

193. Using microarray analysis, detected that up to _____were significantly induced by drought in two Mediterranean pine species.....**113 genes**

194. To date, one of the most extensive programmers to develop trees with resistance to insect pests in temperate regions is in…… **British Columbia.**

195. The act of moving plants by human to different habitat to experience different environment conditions is term as…… **Assisted migration**

196. Assisted migration may be undertaken…..**all given (over long distances, current range limit of particular genotypes and populations, or within the existing range.**

TOPIC NO#107

197. Seed orchards, clone banks and clonal archives are examples of _____conservation units…… **static ex situ**

198. _____ensures the success of species in environments that are highly variable and subject to change…… **Genetic diversity**

TOPIC NO#108-109

199. _____will be fine for cutting tree in 1992 act ……**5000 rupees**

200. The Wild Birds and Animals Protection Act…… **1912**

TOPIC NO#110

201. Convention on biological diversity…… **1992**

TOPIC NO#112

202. Nagoya protocol ratified by _____ partie……..**97**

TOPIC NO#113

203. Cartagena protocol was adopted on ……**June 2001 in Cartagena, Spain.**

204. Cartagena protocols not cover products derived from……**LMOs**

205. Cartagena protocols deals with……**biosafety**

TOPIC NO#117

206. The Pakistan Animal Quarantine Ordinance…….. **1979**

TOPIC NO#118

207. (IABGR) stand for ……..**Institute of Agricultural Biotechnology and Genetic Resources**

208. Number of lab in plant genetic resource in Pakistan…… **six**

TOPIC NO#120

209. Greater efforts are needed to estimate the full value of……. **PGRFA,**

210. In plant tissue culture what is the term ORGANOGENESIS means…… **Formation of root and shoot from callus culture(all)**

211. Plant are totipotent because……**A single cell could become a complete plant**

212. The type of coiling in DNA is …….. **right-handed**

213. ____is not generally seen in biodiversity hotspots……**Lesser interspecific competition**

214. ____ Factor/factors can affect the services of invertebrates to the environment…….. **ALL Given**

A. BEST OF LUCK

MAAH WARRIORS